Roll No. ....

TCE–601

B. TECH. (CIVIL ENGG.) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

ENVIRONMENTAL ENGINEERING—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the factors affecting per capita generation of sewage. (CO1)

(b) BOD of a sewage incubated for 3 days at 35°C was found to be 180 mg/L. Find the value of 5-day 20°C BOD. Assume *k* (base 10) at 20°C as 0.12 per day. (CO1)

(c) Explain average flow, dry weather flow and maximum flow and factors affecting each. Also provide the mathematical relationship between them. (CO1)

2. (a) A city with a population of 100,000 has an area of 50 km^2. Rate of water supply is 110 litres per capita per day of which 80% turns into sewer. The average run-off coefficient is 0.5 and intensity of rainfall is 14.5 mm/hr. Estimate the quantity of combined sewage. Take peak factor as 2.5. (CO2)

   (b) Derive the first stage BOD equation and ultimate BOD. (CO2)

   (c) A sewer has to be designed considering both minimum velocity and maximum velocity of flows. State true or false and justify the answer. (CO2)

3. (a) Design a circular primary sedimentation tank to treat an average sewage flow of 5000 m^3/day, suitably assuming the design criteria. Draw a neat sketch of the designed tank. (CO3)

   (b) Design a rectangular sedimentation tank for treating 12 MLD adopting L : B ratio as 2.5 and overflow rate 40 m^3/m^2/day Assume Detention time as 2 hours. (CO3)

   (c) State the objectives of treatment processes. What are the treatment processes ? Explain. Discuss the various types of screens adopted in sewage treatment with neat sketch. (CO3)

4. (a) Write in detail about the UASB reactor with neat sketch, advantages and disadvantages. (CO4)

   (b) Discuss the operational principles of activated sludge process. (CO4)

   (c) Determine the size of a high-rate trickling filter for the following data :

   Sewage flow = 4.5 million litres per day (CO4)

   Recirculation ratio = 1.5

   BOD for raw sewage = 230 mg/L

   BOO removal in PST = 30%

   BOD of treated effluent required =25 mg/I.

5. (a) Name the various actions involved in the self-purification process of a stream and explain briefly. (CO5, CO6)

   (b) Explain the methods available and limitations of land disposal of sewage. (CO5, CO6)

   (c) Explain the various process involved in sludge treatment and disposal. (CO5, CO6)

Roll No. ....

TCE–602

B. TECH. (CE) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

REINFORCED CEMENT CONCRETE—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain with proper reasoning various advantages and disadvantages of prestressed concrete design ? Illustrate various methods of analysis of prestressed concrete member. (CO1)

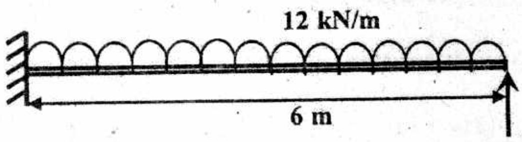
   (b) A rectangular PSC beam given cross section details of 250 × 450 mm size, with 8 m long span is carrying a UDL of 14 kN/m over the entire span and provided with 8-10 mm dia steel wires positioned at 75 mm from bottom fiber of the beam. A net prestress of 850 N/mm^2 is applied on all the wires. Determine the position of the resultant thrust at center of the of the given PSC beam. Assume unit weight of the concrete as 25 kN/m^3. (CO1)

P. T. O.

(c) Explain and derive equations for any *five* different types of losses in prestressed concrete members. (CO1)

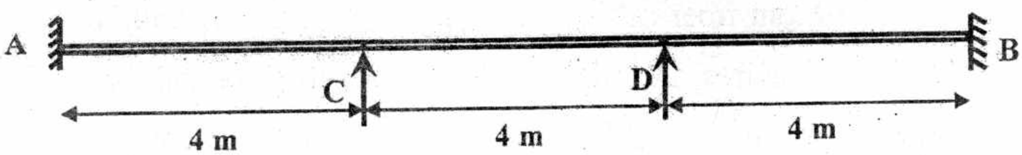
2. (a) State and explain your understanding with proper reasoning's behind the assumptions and guidelines made to apply moment redistribution for a statically indeterminate structure. (CO2)

(b) Draw the bending moment diagram of a propped cantilever beam carrying characteristic live load of 12 kN/m in addition to dead load of 5 kN/m over entire span of 7 m. Develop the Bending Moment envelop for the beam with 25% moment redistribution for the beam. (CO2)

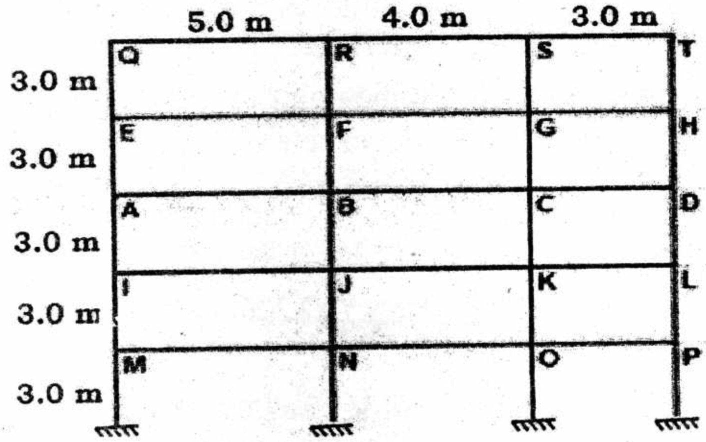


(c) Explain how the concept of plastic hinge and mechanism formation is applied in the concept of moment re-distribution ? Demonstrate the same in a fixed beam carrying a udl of 'w' over its entire span. (CO2)

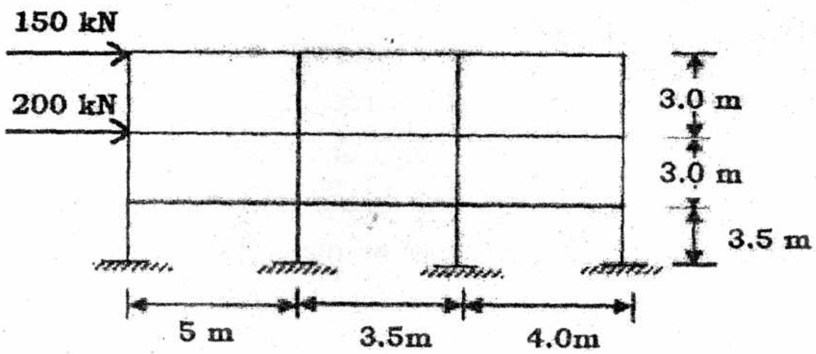
3. (a) Draw the bending moment diagram of the three-span continuous beam carrying characteristic live load of 8 kN/m in addition to its characteristic dead load of 5 kN/m..The length of the all span may be considered as 4 M. Determine the maximum positive bending moment in the middle span. (CO2)



(b) Analyze the member forces for the substitute frame ABCD selected, for Max Negetive BM at Interior support, if the frame memers have same section modulus and load details as given below. Member dead load of 5 kN/m and a live load of 8 kN/m run. (CO3)



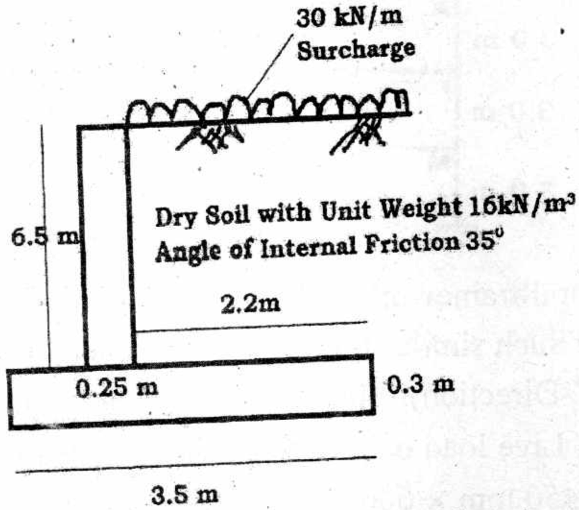
(c) The structural framework (X-Y Plane) for an office building is shown in the Figure. Such similar frames are at a space of 5.0 m in the Transverse direction(Z-Direction). The other data is as follows : Live load on floor = 4 kN/m², Live load on roof = 1.5 kN/m², Beam = 350 mm × 500 mm, Column = 450 mm × 600 mm. Analyze the frame for the axial forces in the bottom most storey columns by using portal method (CO3)



4. (a) Formulate the width of a cantilever retaining wall satisfying the stability against overturning and sliding, whel all other parameters like safe bearing capacity, soil angle of friction, active earth pressure coefficient and soil unit weights are given. (CO4)

(b) Design for counter fort base slab on soil retention side which retains 6m high embankment above ground level. The foundation has to be taken 1.1.m deep. Soil SBC is 200 kN/m². The top of earth retained is horizontal and soil unit wt is 17 kN/m³ with an angle of internal friction 28°. And coefficient of friction between concrete and soil is 0.35. Use M20 and Fe 415 Grade materials. (CO4)

(c) Decide the magnitude and the point of action of the active earth pressure for the given case of cantilever retaining wall ? (CO4)



5. (a) Discuss a comparative salient design considerations for the design of underground, over ground and elevated rectangular water tanks. (CO5)

(b) Design the short side wall a rectangular concrete water tank of size 2.5 × 5 m on the ground, for 5 day storage capacity for a hostel of 20-member residential capacity as per national water consumption norms. A freeboard of 150 mm shall be provided. The wall of the tank is fixed at the base and free at the top. Use M25 and Fe 415 steel. (CO5)

(c) Describe the major components and their important functionality of an Intze tank with the help of a neat sketch. (CO5)

Roll No. ....

TCE–603

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

WATER RESOURCES ENGINEERING—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Obtain an expression for the base width of the elementary profile for no tension condition. (CO1, CO6)

(b) In what way earthquake affect the stability of dam ? Briefly outline the procedure adopted to take in to account the seismic force in the design of a dam. (CO1, CO6)

(c) A concrete gravity dam has maximum reservoir level as 200.0 m, base level as 115.0 m, top level of dam is 205.0 m. The upstream face is vertical and the downstream face batter starts at 199.0 m and is 1.0 H :

1.5 V. There is no tail water and consider full uplift pressure at heel. Calculate : (CO1, CO6)

(i) Maximum vertical stresses at heel and toe.

(ii) Major principal stress and shear stress near the toe

Neglect earthquake and wave action.

2. (a) Discuss briefly the various causes of failure of earthen dam. (CO2, (CO6)

(b) Prove that the most economical section of an arch dam has a central angle of 133°34. (CO2, CO6)

(c) An earth dam made of a homogeneous material has the following data :

Coefficient of permeability of dam material - 5×10^{-4} cm/ sec.

Level of top of dam = 200.0 m.

Level of deepest river bed = 178.0 m

H. F. L. of reservoir = 197.5 m.

Width of the top of dam = 4.5 m.

Upstream slope = 3 : 1.

Downstream slope = 2 : 1.

Horizontal filter of length equal to 25 m is provided inward from the downstream toe of the dam.

Determine the phreatic line for this dam section and the discharge passing through (CO2, CO6)

3. (a) Enumerate the important types of spillway gates. Describe with a neat sketch the construction and working of Tainter gate. (CO3, CO6)

(b) Explain the important relationship between tail water curve and jump height curve in the selection of energy dissipater. (CO3, CO6)

(c) A high spillway is to discharge 20 m^3/sec/m lengths. If the maximum water level is to be 50 m above river bed. Obtain the profile of the spillway crest using USWES method. Take the u/s face as vertical and d/s face of 0.8 H : 1.0 V. (CO3, CO6)

4. (a) State the fundamental difference between Khosla's and Bligh's creep theory for seepage flow below a weir. (CO4)

(b) What are the silt control devices ? Explain their functions. (CO4)

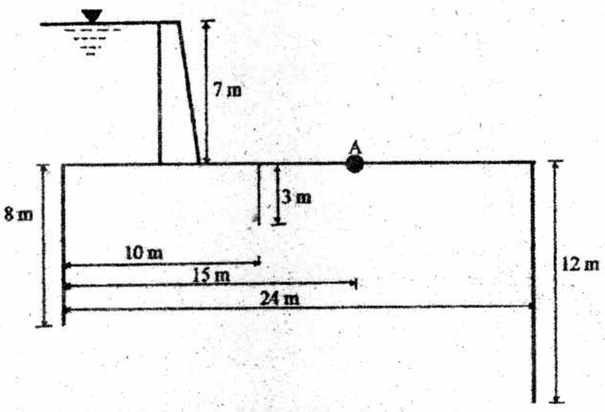
(c) The following figure shows the section of a hydraulic structure on a permeable foundation. Determine :

(i) Average hydraulic gradient according to Bligh's creep theory

(ii) Uplift pressure at A

(iii) The floor thickness required at A

Take S = 2.24



5. (a) What is meant by water-logging ? What are its ill effects ? Describe some anti-water-logging measures. (CO5, CO6)

(b) State the advantages and disadvantages of hydropower generation over thermal and nuclear power generation. (CO5, CO6)

(c) Determine the spacing of the tile drain from the following data :

Average annual rainfall = 1000 mm.

Discharge usually taken is 1% of average annual rainfall in 24 hours

Depth of impervious strata from land surface = 11 m.

Highest position of water table below land surface = 1.5 m.

Depth of drain = 1.8 m below land surface.

Permeability of soil = 6×10^{-6} m/s. (CO5, CO6)

100

Roll No. ....

TCE–604

B. TECH. (CIVIL ENGG.) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

QUANTITY ESTIMATION AND COSTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the purpose and importance of estimates in civil engineering works. (CO1)

   (b) Explain cubical content rate estimate and, preliminary estimate. (CO1)

   (c) Prepare a preliminary estimate of a four storied office building having total carpet area of 1500 sq.m for obtaining the administrative approval of the government, given the following data. It may be assumed that 30% of the built up area will be taken up by corridors, verandah, lavatories, staircase etc. Plinth area rate is ₹ 925/- per sqm. Consider the following given charges : (CO2)

   (i) Extra for special architectural treatment 0.5% of building cost.

   (ii) Extra due to deeper foundation at site 2.5% of building cost.

   (iii) Extra for water supply and sanitary installation 7% building cost

   (iv) Extra for internal electrical installation 11.5% of building cost.

(v) Extra for other services 6% of building cost.

(vi) Contingencies - 2.5%

(vii) Supervision charges - 10 %

2. (a) Calculate the quantities of material required for the following items of work : (CO2)

   (i) Cement plaster 15 mm thick cement mortar 1 : 4 for 600 m^2

   (ii) Cement concrete 1 : 3 : 6 for 10 cum

   (b) Explain Cost of materials, Lead statement, Cost of labour and Contractor's profit. (CO2)

   (c) Explain the factors on which rates of items depend. (CO2)

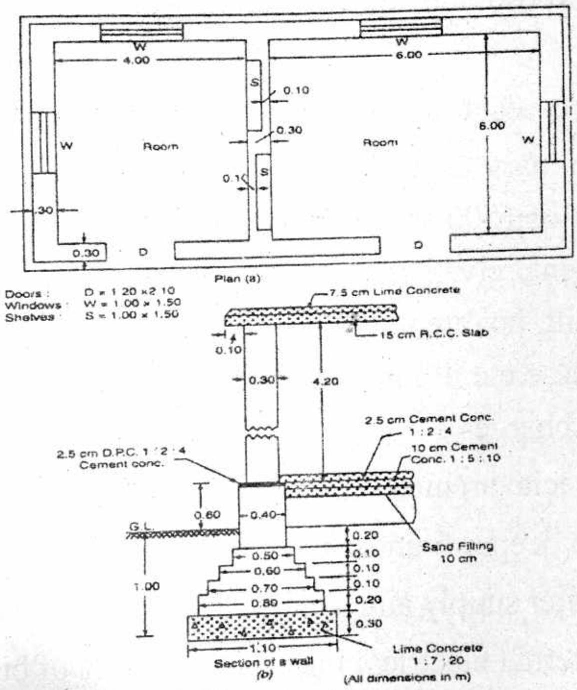
3. (a) Estimate by center line method the quantities of the following items of a residential building shown below using Center line method : (CO3)

   (i) Earth work in excavation in foundation

   (ii) Lime concrete in foundation (CO3)

   (iii) 1 class brickwork in (1 : 6) cement sand mortar in foundation and upto to plinth

   (iv) DPC of 2.5 cm thick of cement concrete 1 : 2 : 4



(b) Explain mid-sectional area method for earthwork in roads. (CO3)

(c) Prepare a detailed estimate for earthwork for a portion of a road from the following data : (CO3)

Dist. (in m)	R. L. of ground
0	114.50
100	114.75
200	115.25
300	115.20
400	116.10
500	116.85
600	118.00
700	118.25
800	118.10
900	117.80
1000	117.75
1100	117.90
1200	119.50

R. L. of formation 115 Upward gradient 1 in 200 upto 600 m →←

Downward gradient 1 in 400 m.

Formation width of road is 10 m side slope 2 : 1 in banking and 1½ : 1 in cutting. Adopt suitable rates.

4. (a) Explain various percentages for different services in building.(CO4)

   (b) Explain the specification and classify them. (CO4)

   (c) Describe overhead charges and work charge establishment. (CO4)

5. (a) Explain Valuation and purpose of valuation. (CO5)

   (b) Describe Sinking fund, Depreciation. (CO5)

   (c) Explain all the methods of calculating depreciation. (CO5)

Roll No. ....

TCE–605

B. TECH. (CIVIL ENGG.) (SIXTH SEMESTER) END SEMESTER EXAMINATION, June, 2023

TRANSPORTATION ENGINEERING-I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are ***twenty***.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the specific functions and compare characteristic features of different modes of transportation. (CO1)

(b) Calculate the super-elevation required on a road curve of 240 m radius with mixed traffic condition. The design speed is 80 kmph. The coefficient of friction is 0.15. The road is passing through rolling terrain. (CO1)

(c) A car travelling at 22.22 m/sec is overtaking another car moving at 16.67 m/sec on a two lane undivided highway. Assuming an acceleration of the overtaking car as 0.7 m/sec^2. Calculate minimum overtaking sight distance. (CO1)

2. (a) Differentiate between bitumen and tar. Discuss the various tests conducted on Bitumen in road construction. (CO2)

(b) Explain the desirable properties of soil as a highway material. (CO2)

P. T. O.

(c) List various tests to be carried on aggregate sample. Explain any *two* tests in detail. (CO2)

3. (a) Explain advantages and disadvantages of rotary intersections. (CO3)

(b) Explain different methods of origin and destination study. (CO3)

(c) Explain regulatory sign, warning sign and informatory sign with neat sketches. Discuss the advantages and disadvantages of traffic signals. (CO3)

4. (a) Explain the difference between flexible and rigid pavements. (CO4)

(b) Explain the difference factors affecting flexible pavement. (CO4)

(c) Design the pavement for a new road in plain terrain with a two-lane single carriageway; given the following data :

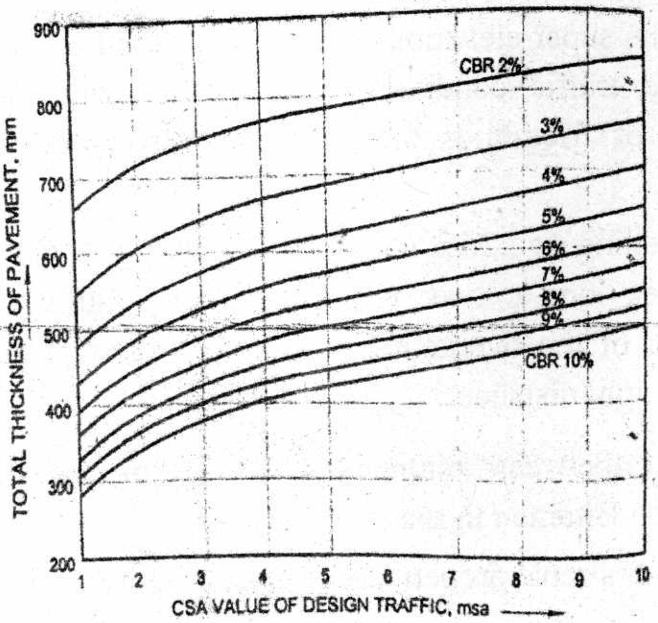
Initial traffic in a year of completion of construction work in both directions = 500 CVPD

Traffic growth rate per annum = 7.5 percent

Design life = 12 years

Vehicle damage factor = 3.5 (standard axles per commercial vehicle)

Design CBR value of sub-grade soil = 4 % (CO4)



5. (a) Explain the different components of an airport Layout. (CO5)

   (b) Explain wind rose diagram. What is its utility and its types ? Explain wind rose diagram type I with neat sketches. (CO5)

   (c) The length of runway under standard conditions is 2100 m. The airport is to be provided at elevation of 410 m above mean sea level.. The airport reference temperature is 32°C. If the runway is to be constructed with an effective gradient of 0.218 percent, determine the corrected runway length. (CO5)